```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
tips = sns.load_dataset("tips")
```

```
tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

Next steps: Generate code with `tips` New interactive sheet

```
flight = sns.load_dataset("flights")
```

```
flight.head()
```

	year	month	passengers
0	1949	Jan	112
1	1949	Feb	118
2	1949	Mar	132
3	1949	Apr	129
4	1949	May	121

Next steps: Generate code with `flight` New interactive sheet

```
iris1 = sns.load_dataset("iris")
```

```
iris1.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

```
titnic = sns.load_dataset("titanic")
```

```
titnic.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	ad
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

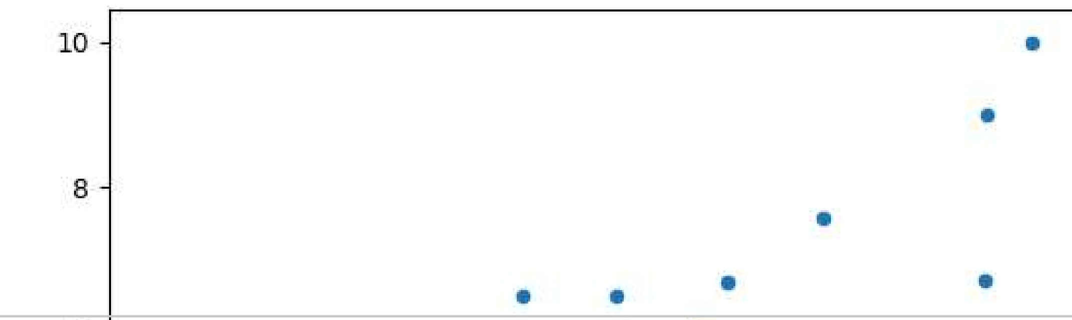
Next steps: Generate code with titnic New interactive sheet

```
tips.columns
```

```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'],
dtype='object')
```

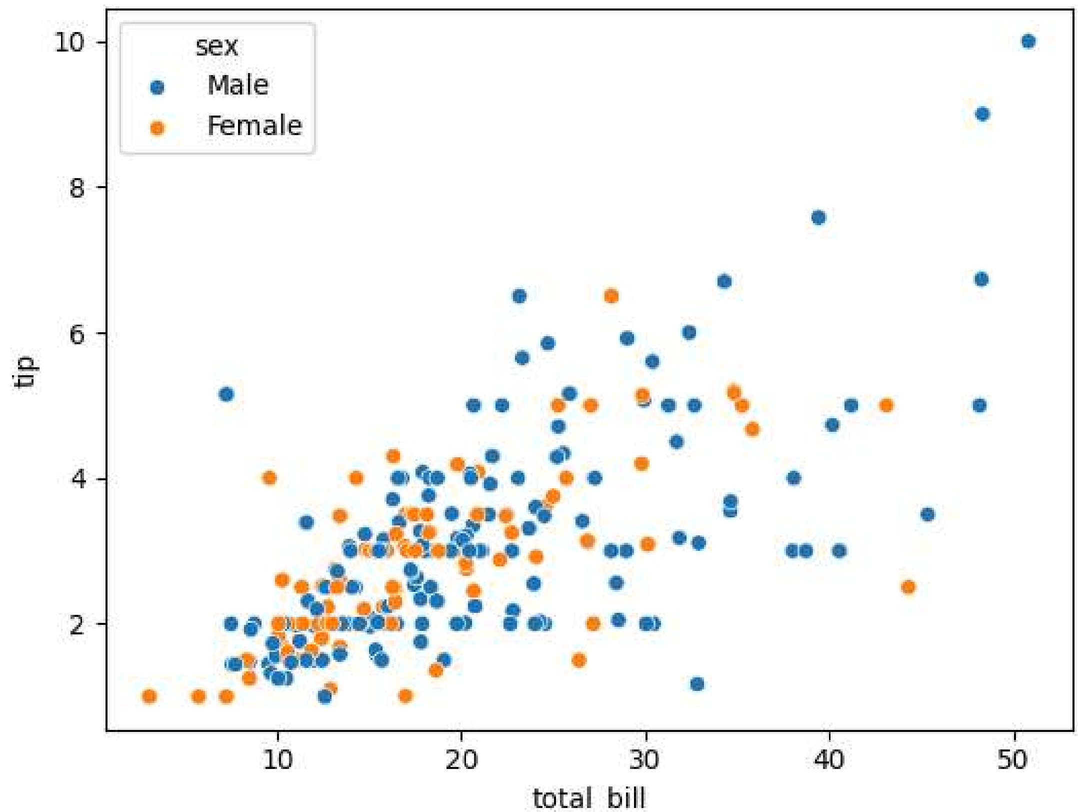
```
sns.scatterplot(x = tips["total_bill"],y = tips["tip"])
```

<Axes: xlabel='total_bill', ylabel='tip'>



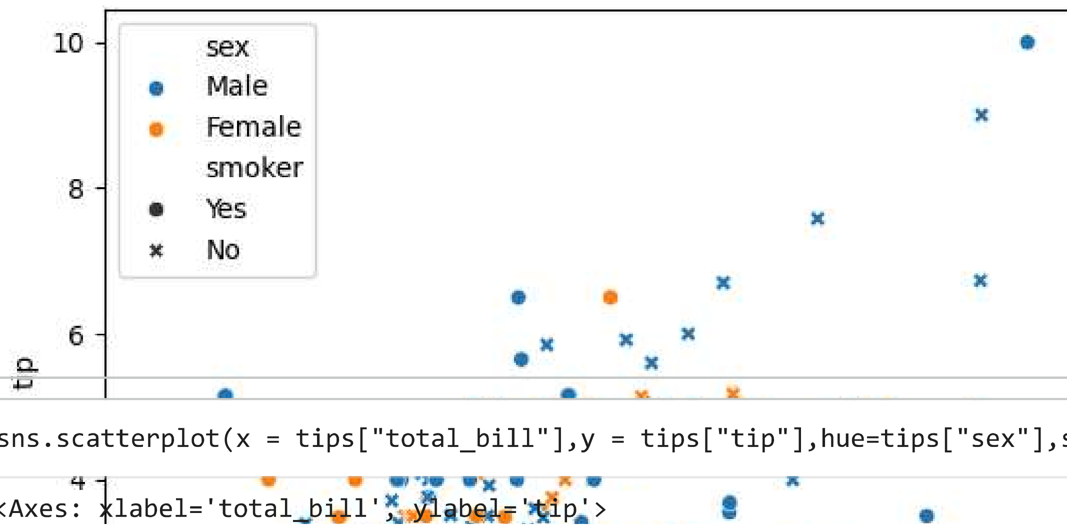

```
sns.scatterplot(x = tips["total_bill"],y = tips["tip"],hue=tips["sex"])
```

<Axes: xlabel='total_bill', ylabel='tip'>



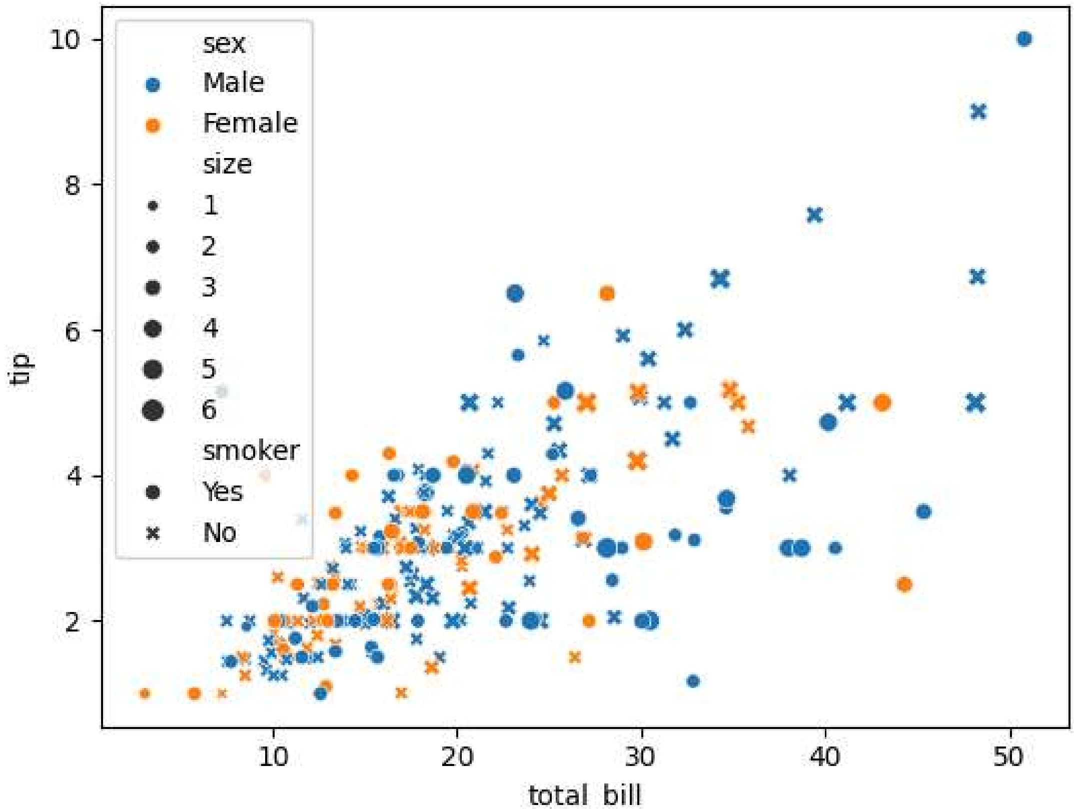

```
sns.scatterplot(x = tips["total_bill"],y = tips["tip"],hue=tips["sex"],style=tips['
```

```
<Axes: xlabel='total_bill', ylabel='tip'>
```




```
sns.scatterplot(x = tips["total_bill"],y = tips["tip"],hue=tips["sex"],style=tips['
```

```
<Axes: xlabel='total_bill', ylabel='tip'>
```

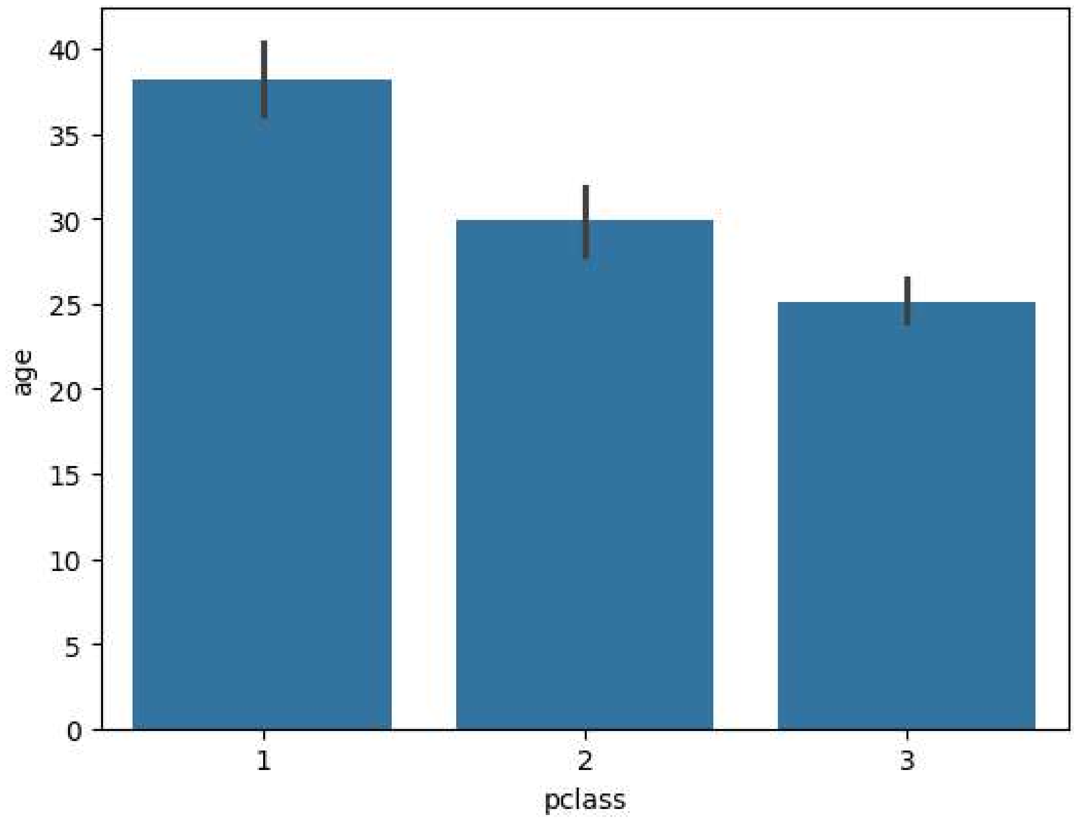



```
titnic.columns
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

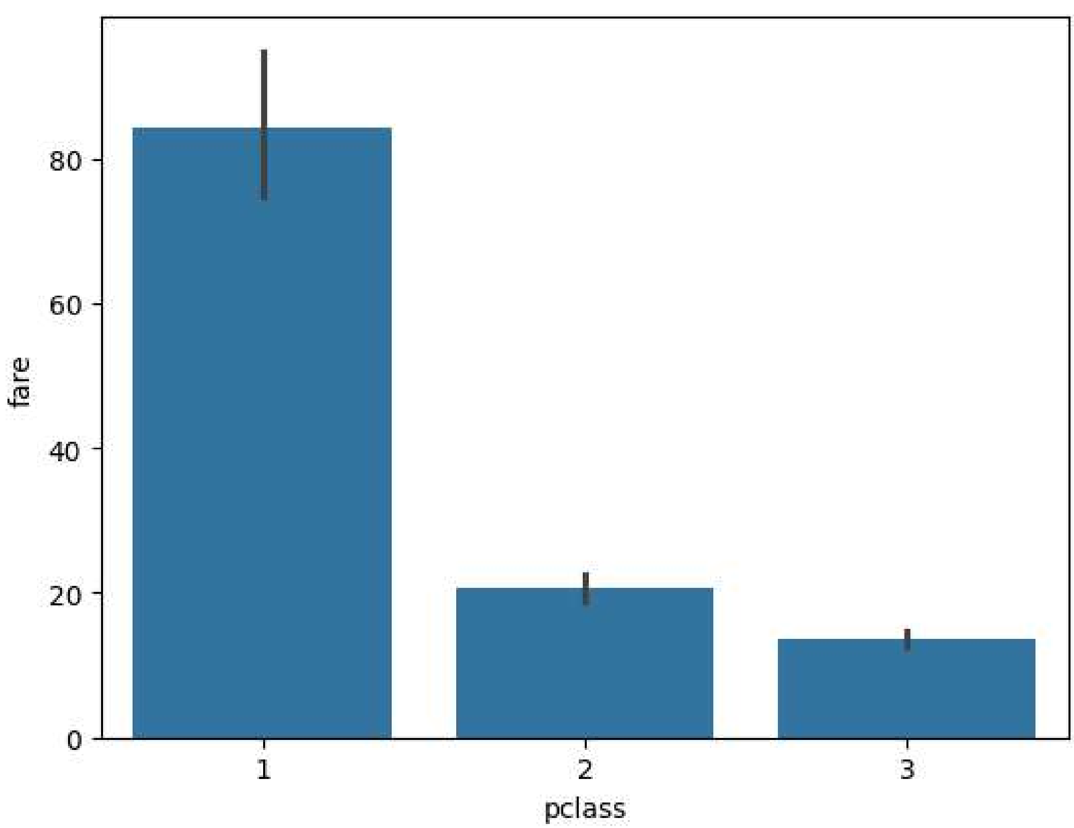
```
sns.barplot(data=titnic, x="pclass", y="age")
```

```
<Axes: xlabel='pclass', ylabel='age'>
```



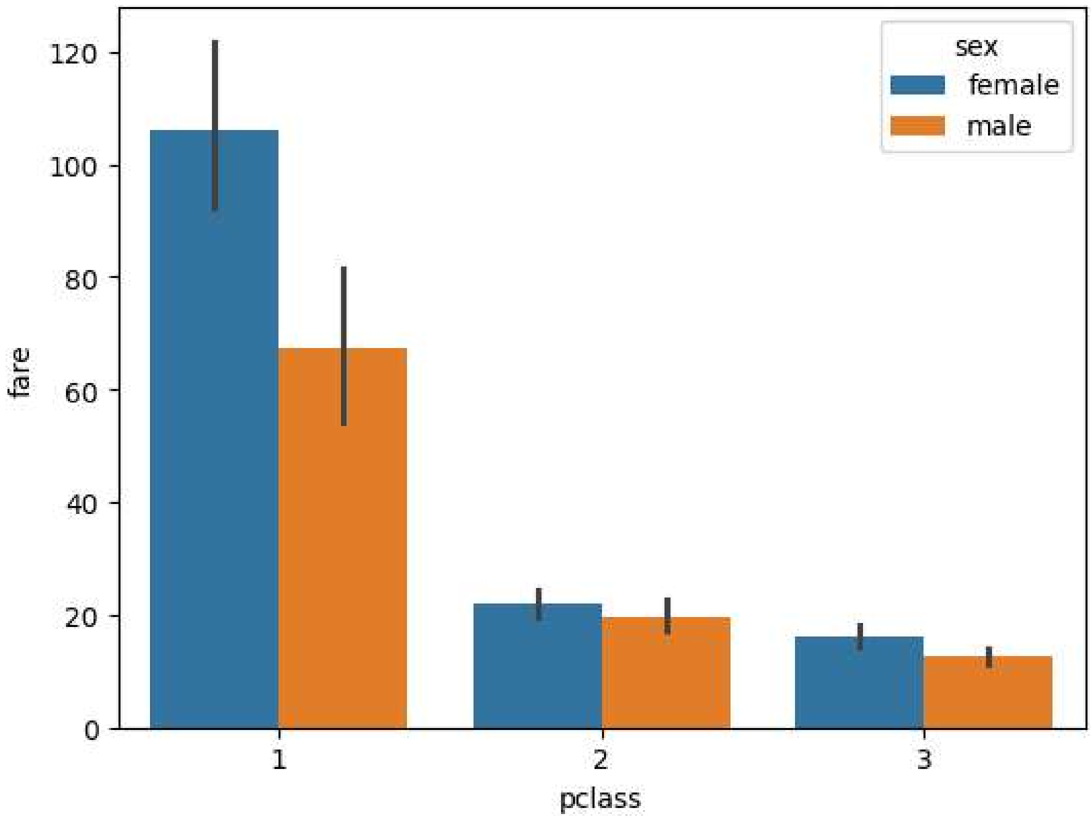

```
sns.barplot(data=titnic, x="pclass", y="fare")
```

```
<Axes: xlabel='pclass', ylabel='fare'>
```



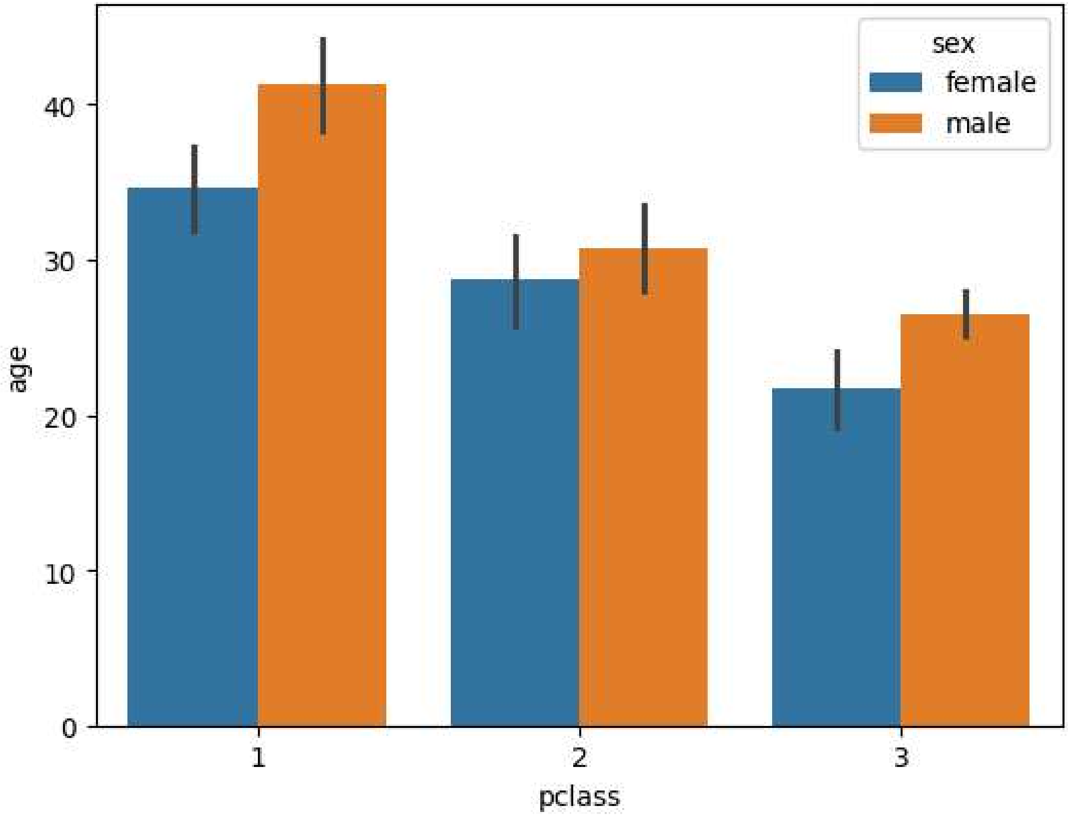
```
sns.barplot(data=titnic, x="pclass", y="fare",hue="sex")
```

```
<Axes: xlabel='pclass', ylabel='fare'>
```



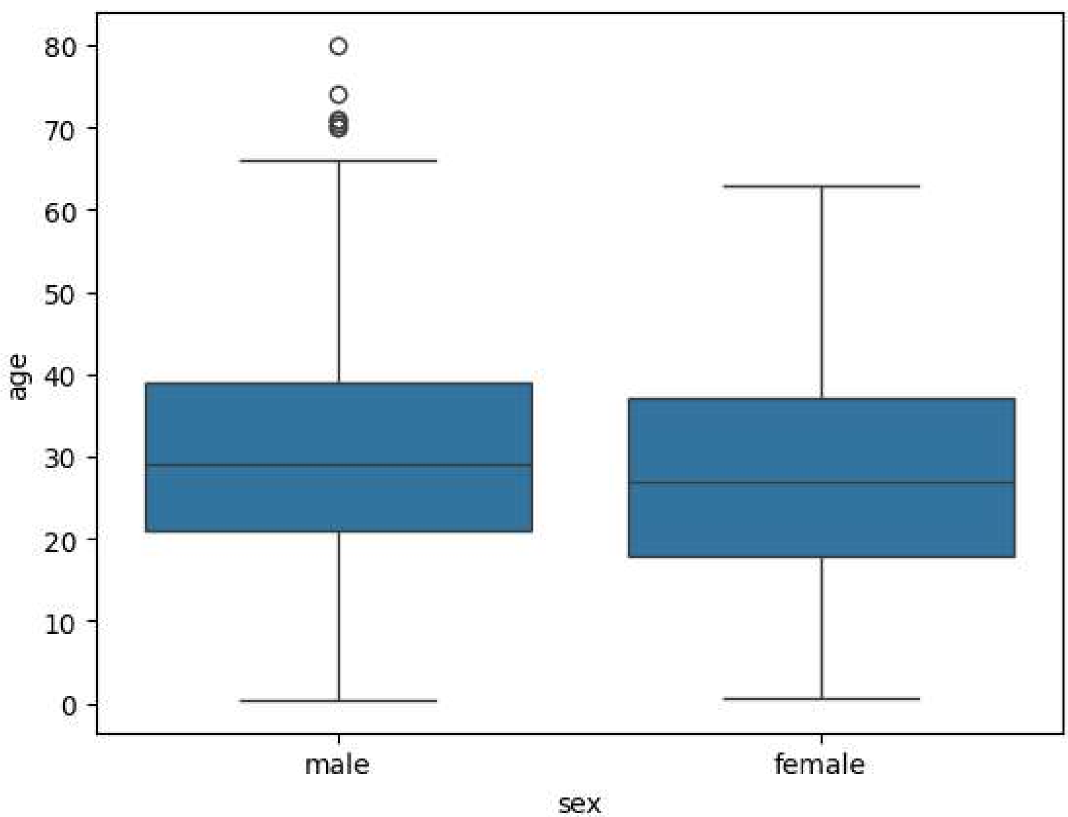

```
sns.barplot(data=titnic, x="pclass", y="age",hue="sex")
```

```
<Axes: xlabel='pclass', ylabel='age'>
```



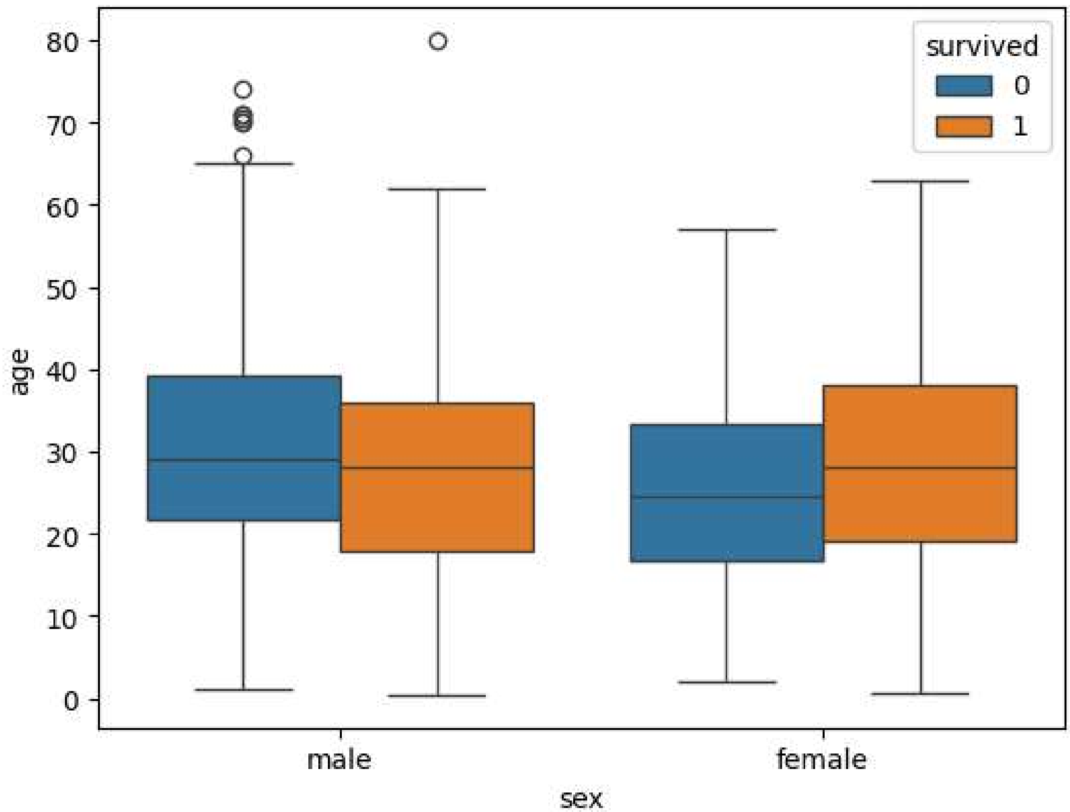

```
sns.boxplot(data = titnic,x="sex",y="age")
```

```
<Axes: xlabel='sex', ylabel='age'>
```



```
sns.boxplot(data = titnic,x="sex",y="age",hue="survived")
```

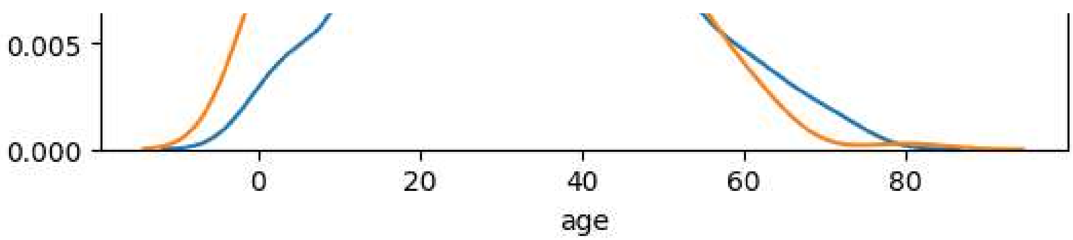
```
<Axes: xlabel='sex', ylabel='age'>
```



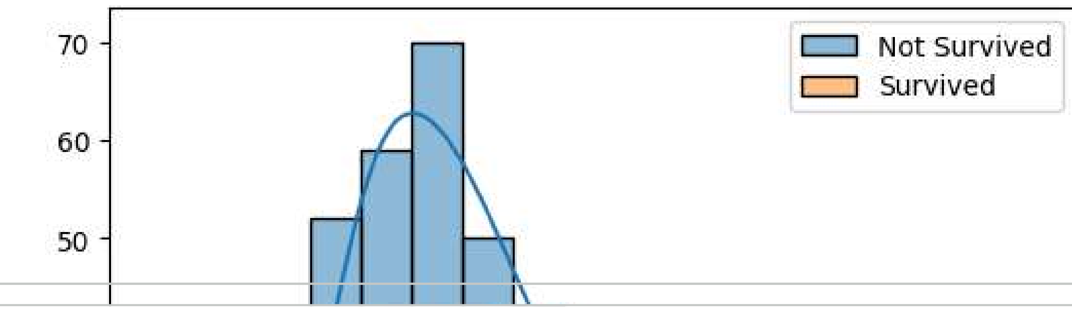

```
sns.distplot(titnic[titnic["survived"]==0]["age"],hist=False)
sns.distplot(titnic[titnic["survived"]==1]["age"],hist=False)
```

```
/tmp/ipython-input-3436488348.py:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `kdeplot` (an axes-level function for kernel density plots)

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(titnic[titnic["survived"]==0]["age"],hist=False)
/tmp/ipython-input-3436488348.py:2: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `kdeplot` (an axes-level function for kernel density plots)

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(titnic[titnic["survived"]==1]["age"],hist=False)
<Axes: xlabel='age', ylabel='Density'>
```

```python
sns.histplot(
    data=titnic[titnic["survived"] == 0],
    x="age",
    kde=True,
    label="Not Survived"
)

sns.histplot(
    data=titnic[titnic["survived"] == 1],
    x="age",
    kde=True,
    label="Survived"
)
plt.legend()
```



```
<matplotlib.legend.Legend at 0x7b470288e090>
```




```
titnic.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	ad
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

age

Next steps: Generate code with titnic New interactive sheet

```
titnic.columns
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

```
pd.crosstab(titnic["pclass"],titnic["survived"])
```

survived	0	1
pclass		
1	80	136
2	97	87
3	372	119

```
pd.crosstab(titnic["pclass"],titnic["survived"]).reset_index()
```

survived	pclass	0	1
0	1	80	136

```
pd.crosstab(titnic["pclass"],titnic["survived"]).rename(columns={0 : "Not survied",
```

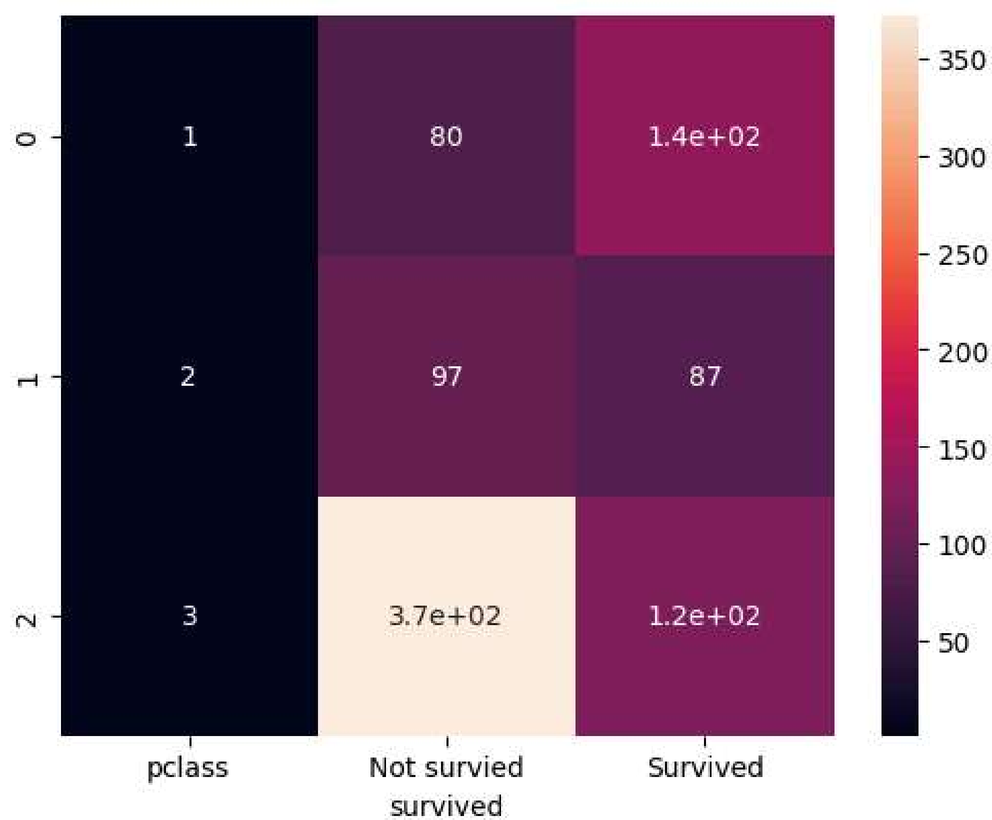
survived	pclass	Not survied	Survived
0	1	80	136
1	2	97	87
2	3	372	119

```
pd.crosstab(titnic["pclass"],titnic["survived"]).rename(columns={0 : "Not survied'
```

survived pclass	Not survied	Survived
1	80	136
2	97	87
3	372	119

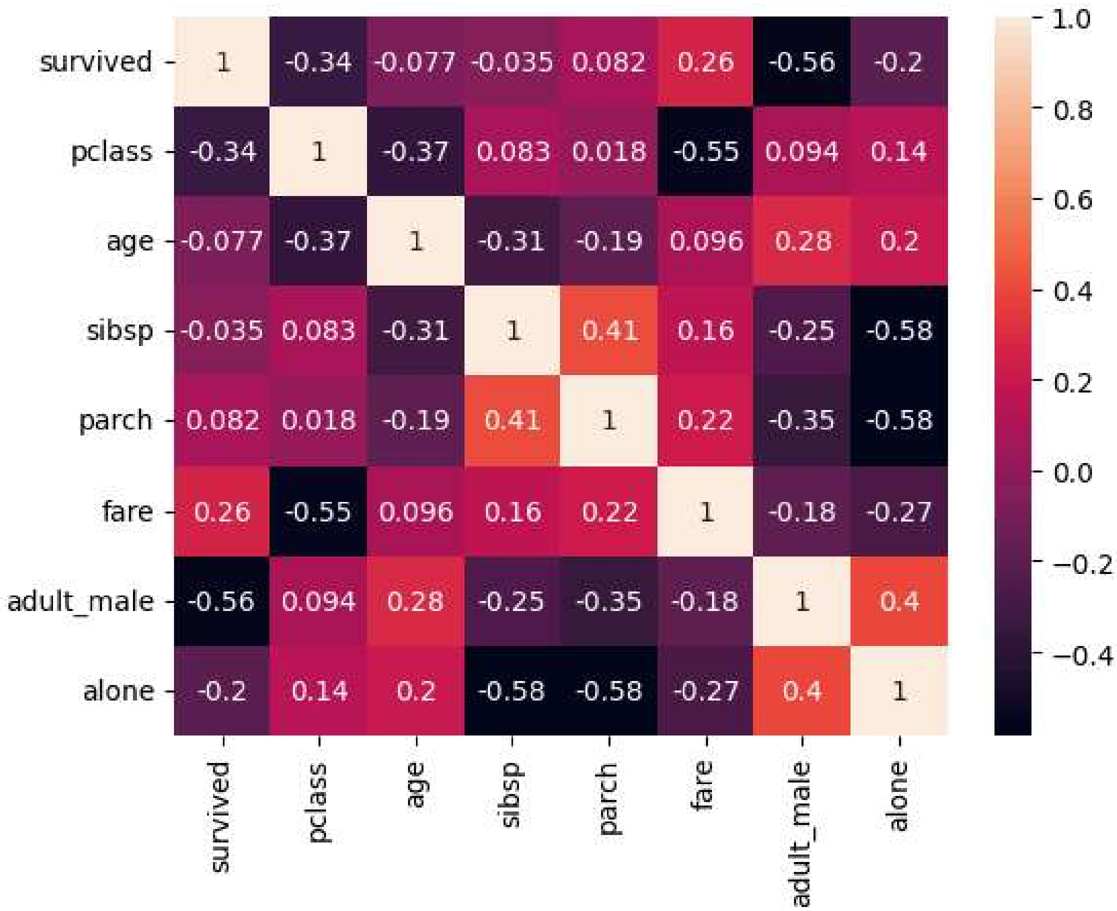
```
sns.heatmap(pd.crosstab(titnic["pclass"],titnic["survived"]).rename(columns={0 : "N
```

```
<Axes: xlabel='survived'>
```



```
corr = titnic.corr(numeric_only=True)
sns.heatmap(corr,annot=True)
```

<Axes: >




```
titnic.groupby("pclass")["survived"].mean()
```

	survived
pclass	
1	0.629630
2	0.472826
3	0.242363

dtype: float64

Double-click (or enter) to edit

```
round(titnic.groupby("pclass")["survived"].mean()*100,2)
```

pclass	survived
1	62.96
2	47.28
3	24.24

dtype: float64

```
round(titnic.groupby("pclass")["survived"].mean()*100,2).astype(int).astype("str")+
```

pclass	survived
1	62%
2	47%
3	24%

dtype: object

```
round(titnic.groupby("pclass")["survived"].mean()*100,2).astype(int)
```

pclass	survived
1	62
2	47
3	24

dtype: int64

```
round(titnic.groupby("pclass")["survived"].mean()*100,2).astype(int).astype("str")
```

pclass	survived
1	62
2	47
3	24

dtype: object

```
round(titnic.groupby("embark_town")["survived"].mean()*100,2).astype(int).astype('
```

	survived
embark_town	
Cherbourg	55%
Queenstown	38%
Southampton	33%

dtype: object

```
titnic.columns
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_town',
       'alive', 'alone'],
      dtype='object')
```

```
round(titnic.groupby("sibsp")["survived"].mean()*100,2).astype(int).astype("str")+"
```

	survived
sibsp	
0	34%
1	53%
2	46%
3	25%
4	16%
5	0%
8	0%

dtype: object

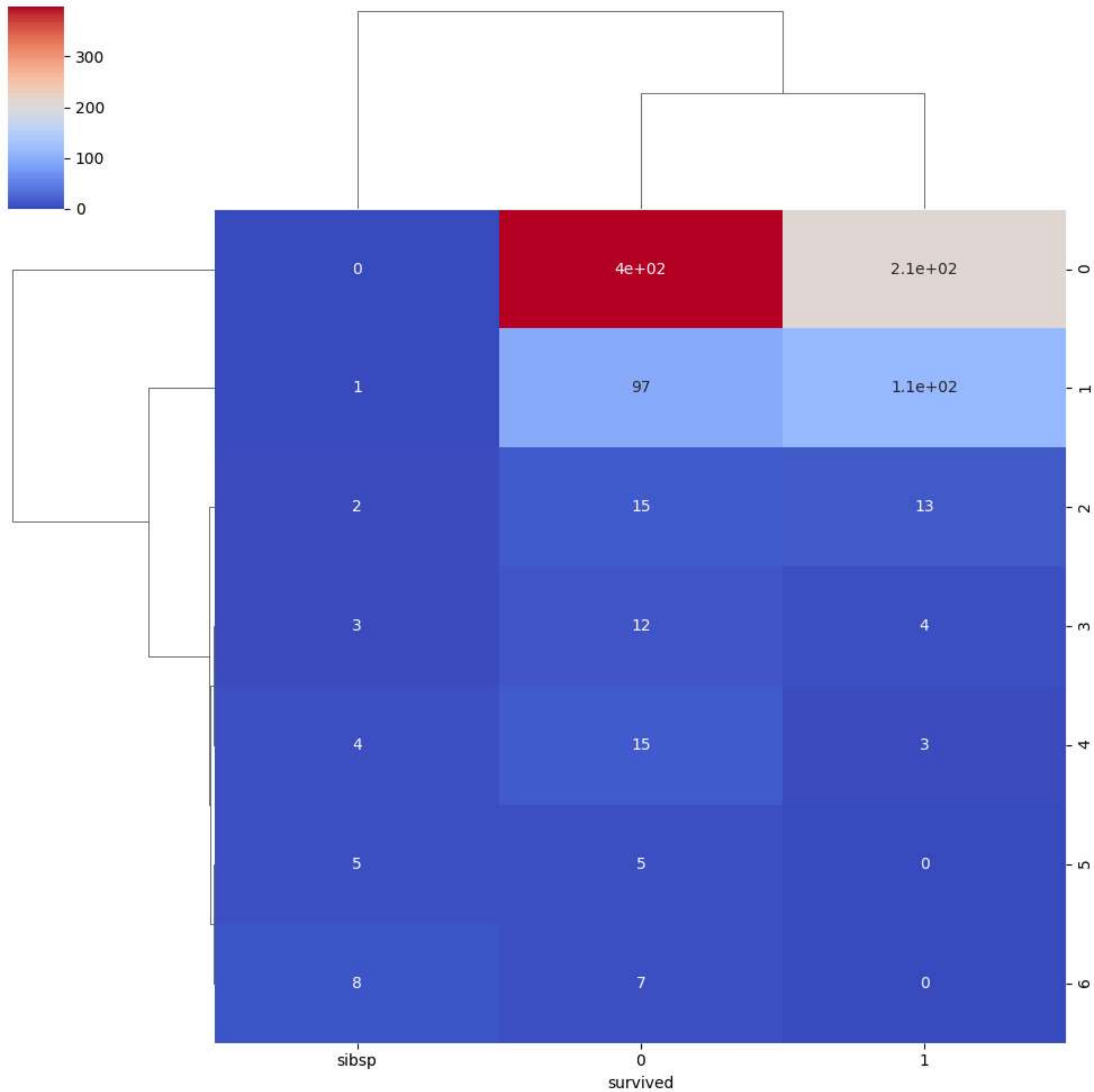
```
ct = pd.crosstab(titnic["sibsp"],titnic["survived"]).reset_index()
ct
```

survived	sibsp	0	1
0	0	398	210
1	1	97	112
2	2	15	13
3	3	12	4
5	5	5	0

```
sns.clustermap(ct,cmap="coolwarm", annot=True)
```

<seaborn.matrix.ClusterGrid at 0x7b470150e690>

Next



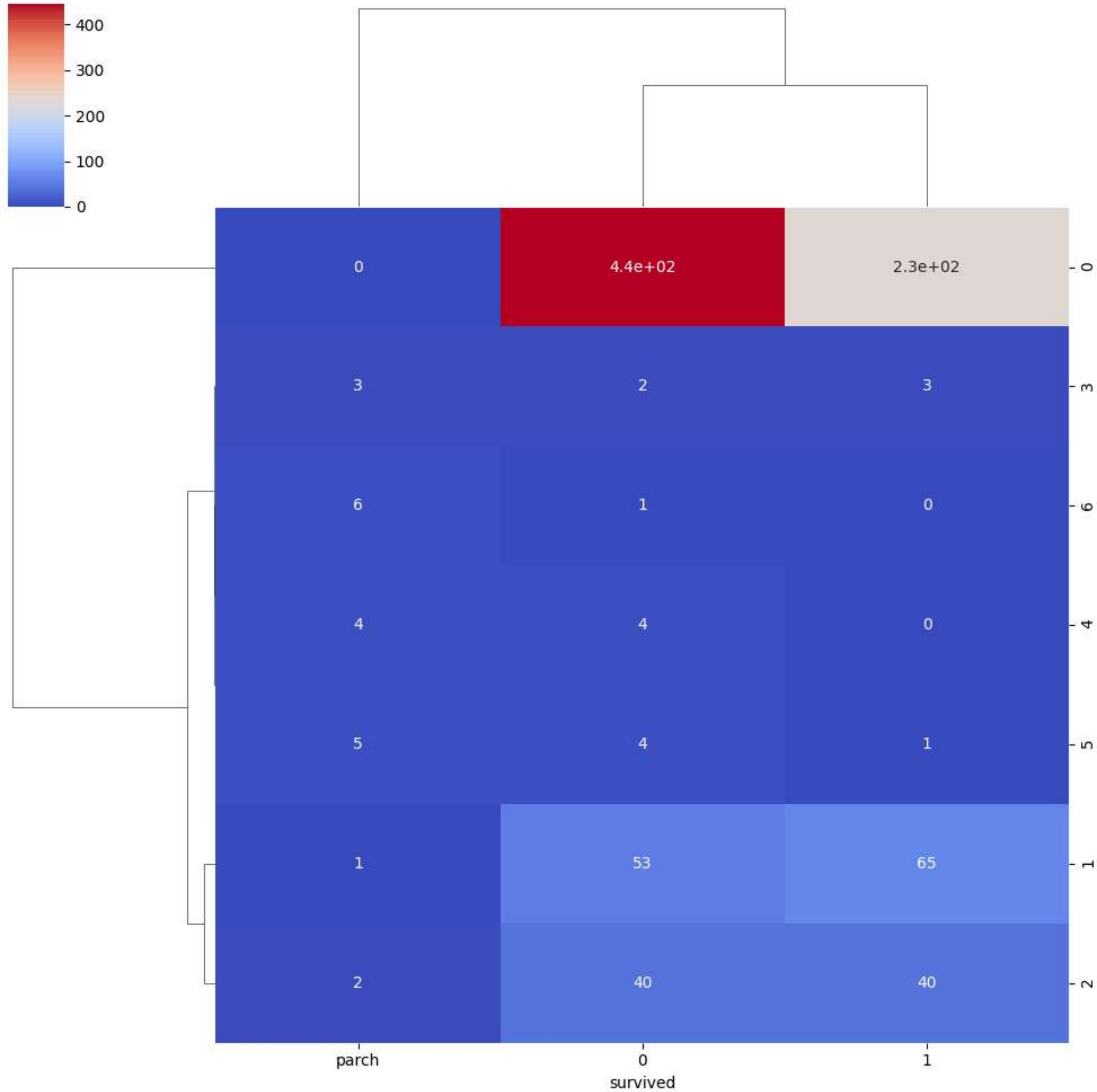

```
ct = pd.crosstab(titnic["parch"],titnic["survived"]).reset_index()
ct
```

survived	parch	0	1
0	0	445	233
1	1	53	65
2	2	40	40
3	3	2	3
4	4	4	0
5	5	4	1
6	6	1	0

Next steps: Generate code with ct New interactive sheet

```
sns.clustermap(ct,cmap="coolwarm", annot=True)
```

```
<seaborn.matrix.ClusterGrid at 0x7b47014a72c0>
```



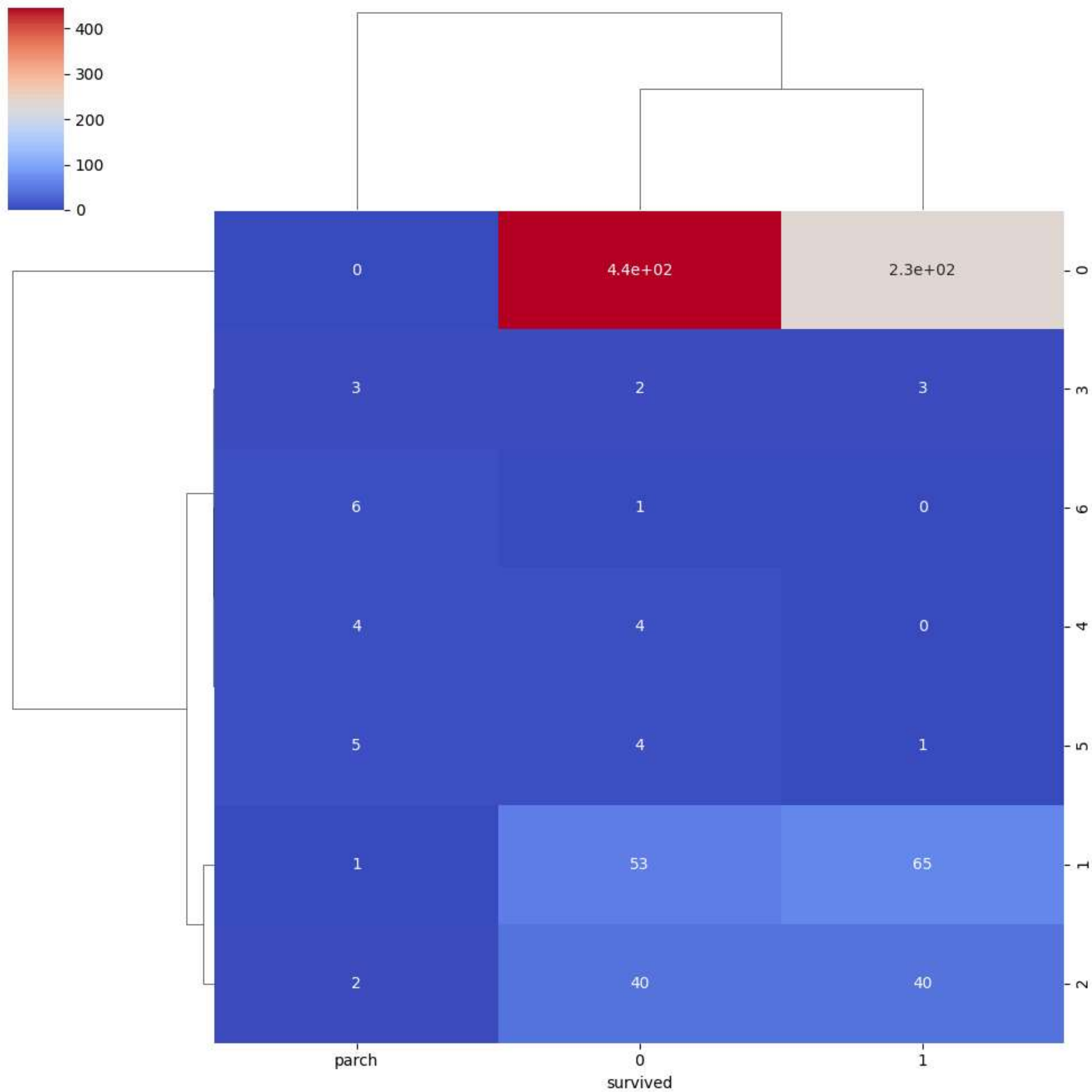

```
ct2 = pd.crosstab(titnic["parch"],titnic["survived"]).reset_index()
ct2
```

survived	parch	0	1
0	0	445	233
1	1	53	65
2	2	40	40
3	3	2	3
4	4	4	0
5	5	4	1
6	6	1	0

Next steps: Generate code with ct2 New interactive sheet

```
sns.clustermap(ct2,cmap="coolwarm", annot=True)
```

```
<seaborn.matrix.ClusterGrid at 0x7b470134d010>
```



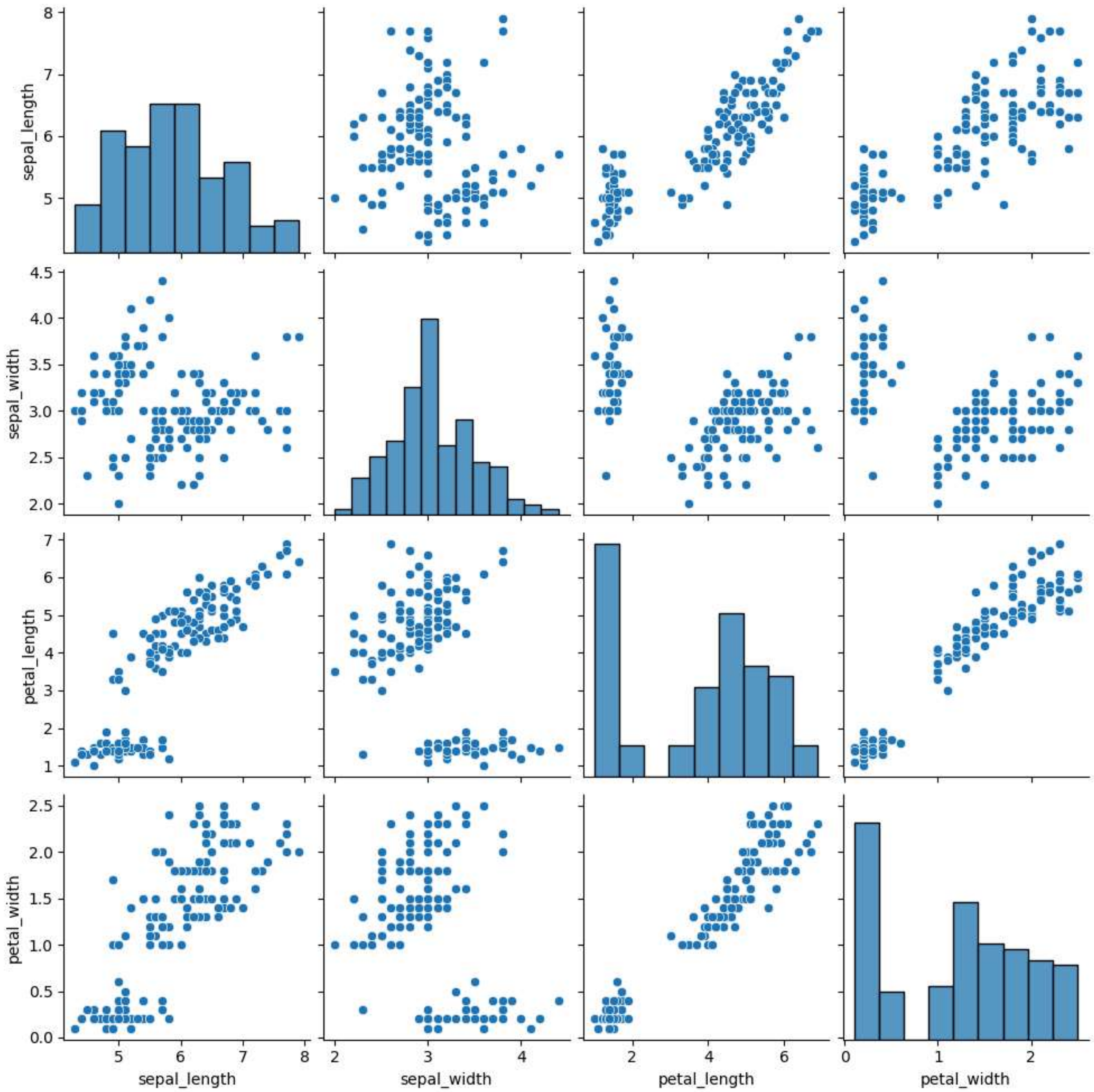

```
iris1.head()
```

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa

Next steps: Generate code with `iris1` New interactive sheet

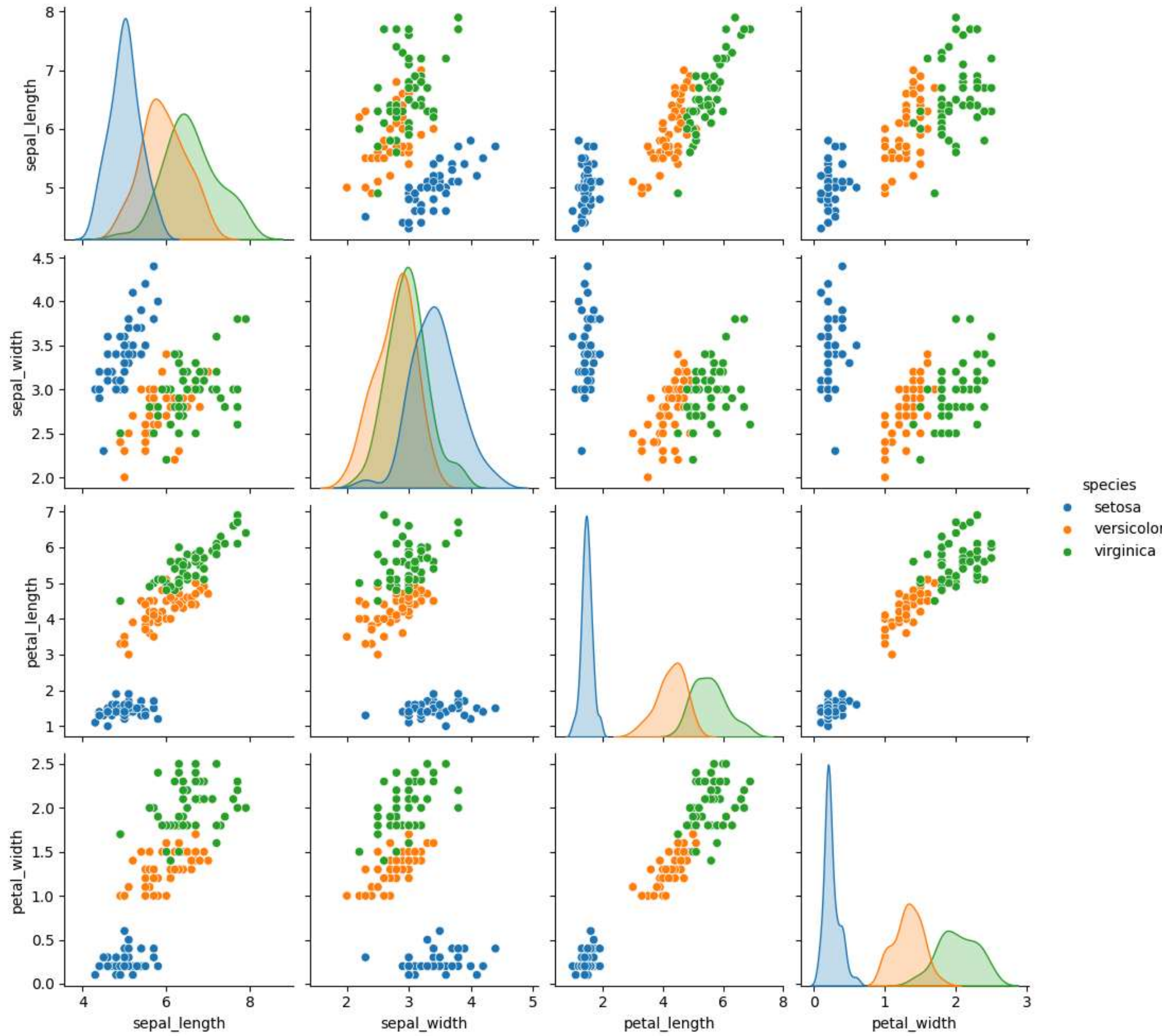
```
sns.pairplot(iris1)
```

```
<seaborn.axisgrid.PairGrid at 0x7b47012bbc20>
```




```
sns.pairplot(iris1,hue="species")
```

```
<seaborn.axisgrid.PairGrid at 0x7b470120ec00>
```




```
flight.head()
```

	year	month	passengers
0	1949	Jan	112
1	1949	Feb	118
2	1949	Mar	132
3	1949	Apr	129
4	1949	May	121

Next steps: Generate code with flight New interactive sheet

```
flight.groupby("year")["passengers"].sum().sort_values()
```

year	passengers
1949	1520
1950	1676
1951	2042
1952	2364
1953	2700
1954	2867
1955	3408
1956	3939
1957	4421
1958	4572
1959	5140
1960	5714

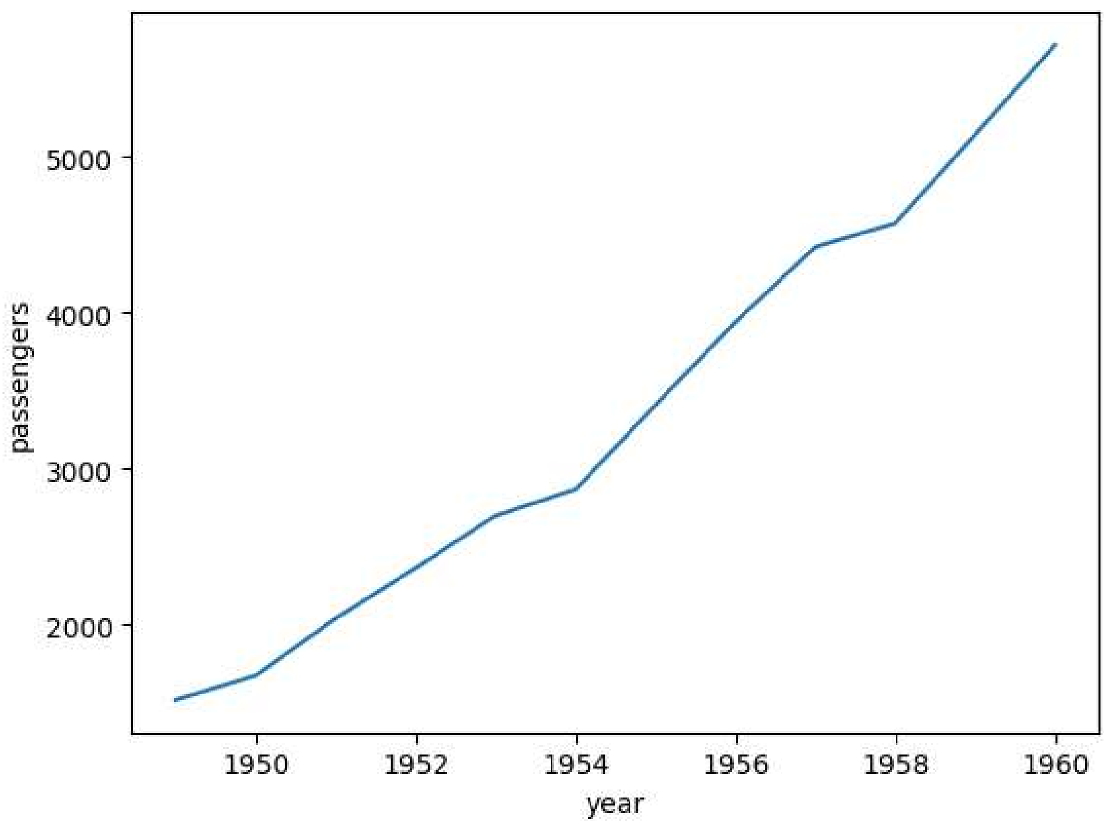
dtype: int64

```
flight.groupby("year")["passengers"].sum().sort_values(ascending=False).reset_inde
```

	year	passengers
0	1960	5714
1	1959	5140
2	1958	4572
3	1957	4421
4	1956	3939
5	1955	3408
6	1954	2867
7	1953	2700
8	1952	2364
9	1951	2042
10	1950	1676
11	1949	1520

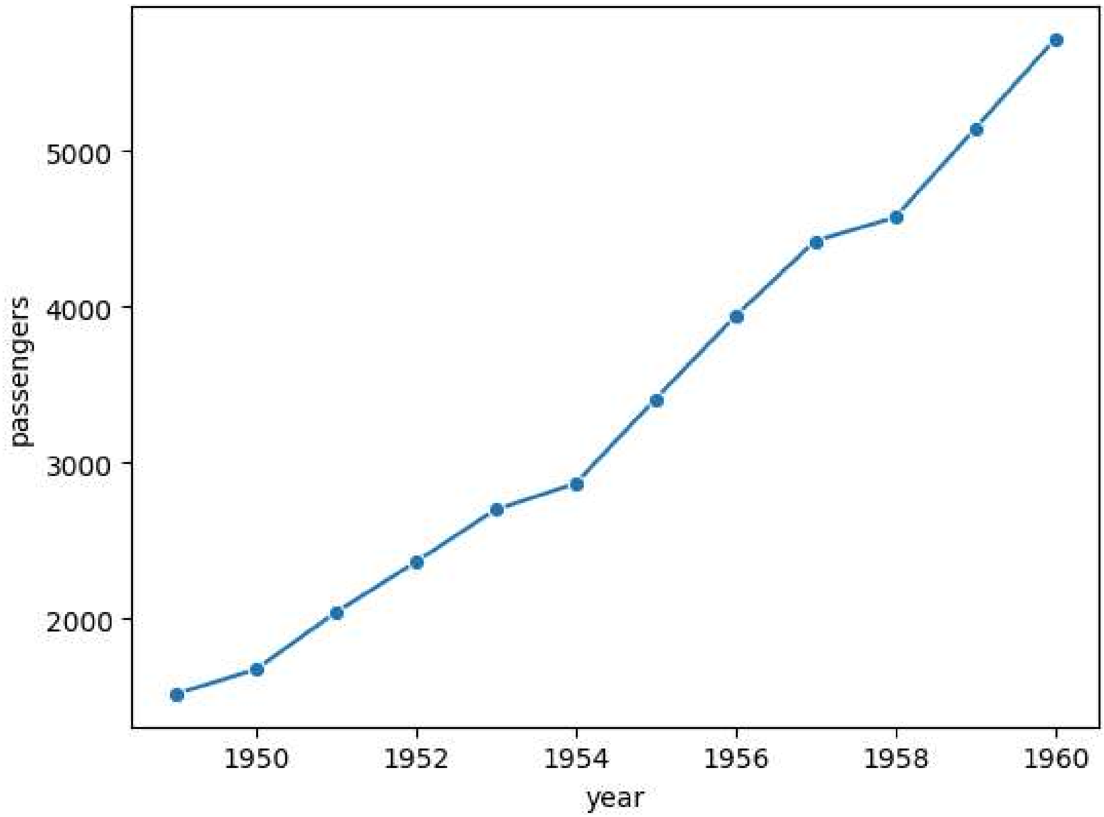
```
sns.lineplot(flight.groupby("year")["passengers"].sum().sort_values(ascending=Fal:
```

```
<Axes: xlabel='year', ylabel='passengers'>
```



```
sns.lineplot(flight.groupby("year")["passengers"].sum().sort_values(ascending=Fal:
```

```
<Axes: xlabel='year', ylabel='passengers'>
```




```
flight.pivot_table(values="passengers",index="month",columns="year")
```

```
/tmp/ipython-input-3253843679.py:1: FutureWarning: The default value of observed=Fa
  flight.pivot_table(values="passengers",index="month",columns="year")
```

year	1949	1950	1951	1952	1953	1954	1955	1956	1957	1958	1959	1960
month												
Jan	112.0	115.0	145.0	171.0	196.0	204.0	242.0	284.0	315.0	340.0	360.0	417.0
Feb	118.0	126.0	150.0	180.0	196.0	188.0	233.0	277.0	301.0	318.0	342.0	391.0
Mar	132.0	141.0	178.0	193.0	236.0	235.0	267.0	317.0	356.0	362.0	406.0	419.0